

PROFILE

Senior Computer Vision Engineer at **Catapult / IMPECT**, leading football computer vision for player tracking and performance analysis. Hands-on experience spans segmentation, classification, tracking evaluation, model deployment, 3x3 basketball research, multi-camera padel tracking, and sports analytics products.

EXPERIENCE

Senior Computer Vision Engineer

Catapult / IMPECT

Aug 2026 - Present

- Lead the technical direction of computer-vision and tracking systems for football movement classification and performance analysis.
- Manage and mentor a small engineering team while remaining hands-on with system design and development.
- Train, optimize, and deploy segmentation and classification models for production football workflows.
- Extend the tracking pipeline to new movements, metrics, and analysis types.
- Evaluate tracking accuracy and build quality metrics to improve reliability across the system.

Researcher and PhD Candidate

Deutsche Sporthochschule Köln

Sep 2024 - Aug 2026

- Developed an automated pipeline for a large-scale 3x3 basketball player-tracking dataset from single-camera broadcast footage.
- Designed game-segment detection, court registration, player detection and tracking, 3D ball localization, and court-coordinate projection methods.
- Addressed identity fragmentation across cuts and occlusions using re-identification, jersey-number recognition, appearance features, and spatiotemporal constraints.

Computer Vision Developer

AiBall

Jun 2023 - Jul 2026

- Developed production multi-camera player and ball tracking for padel, including 3D ball-trajectory reconstruction.
- Designed backend data models and APIs; integrated club-management platforms and automated match recording and processing.
- Built monitoring for data quality, processing pipelines, camera/NVR health, and club performance.

Data Analyst

Ball&Stats;

Oct 2022 - Dec 2023

- Built data extraction, relational storage, and personalized tactical visualizations for Spanish basketball teams.

Lead Data Analyst

Barcelona Dragons

Apr 2022 - Sep 2023

- Built an advanced-statistics warehouse and translated team KPIs into pre-game and tactical analysis with coaching staff.

Data Scientist

Hewlett-Packard (HP)

Sep 2021 - Apr 2023

- Designed complex data flows and applied predictive models to highly unstructured business data.

SELECTED PROJECTS

Computer Vision for Football Performance	Ongoing
• Segmentation, classification, player tracking, movement analysis, and quality evaluation for football at Catapult / IMPECT.	
3x3 Basketball Tracking from Broadcast Video	2024 - 2026
• Automated game segmentation, court registration, player tracking, 3D ball localization, and large-scale dataset development.	
Production Padel Tracking Platform	2023 - 2026
• Multi-camera player and ball tracking, 3D trajectory reconstruction, distributed processing, monitoring, and automated club workflows.	
BUEStats 2.0	Launched
• Web product for Spanish basketball statistics, scraping, scouting, visualization, and player-similarity workflows.	
Team Canada Olympic Rotation Analytics	2024
• Rotation, lineup, and game-dynamics analysis supporting Olympic preparation.	
NBA Player Value and Prediction Lab	2024
• Machine-learning studies on salary efficiency, future-star prediction, and scoring trends.	

EDUCATION

Master in Artificial Intelligence Applied to Sports UCAM - Universidad Católica San Antonio de Murcia	2022 - 2023
Erasmus, Mathematics and Computer Science Technische Universität Darmstadt	2020 - 2021
Bachelor in Data Science and Engineering Universitat Politècnica de Catalunya (UPC)	2017 - 2021

TECHNICAL SKILLS

COMPUTER VISION	Player and ball tracking, segmentation, classification, re-identification, camera calibration, movement classification, and 3D trajectory reconstruction.
ML AND DEPLOYMENT	Model training, evaluation, optimization, inference, deployment, and tracking-quality evaluation.
PROGRAMMING AND DATA	Python, Pandas, SciPy, NumPy, scikit-learn, Keras, SQL, R, MATLAB, data pipelines, relational modeling, and data warehousing.
PRODUCT AND ANALYTICS	Streamlit, Dash, Power BI, Tableau, backend APIs, distributed video processing, and operational monitoring.

LANGUAGES

Catalan - Native/Bilingual | Spanish - Native/Bilingual | English - C1 | German - B2